CI Professional Mentoring and/or Professional Development Plan

Covered Role and Individual Development Plan

If the PI's budget-supported software-engineering role is classified as a CI Professional, this plan applies to that effort; if a research software engineer or programmer is added, the same activities will apply and the final document will name the covered role(s). Within the first month, each covered individual will prepare a development plan with the project science lead and an independent Georgia Tech CI/research-software mentor. The plan will identify technical, scientific, communication, leadership, and career objectives and will be reviewed semiannually.

Mentoring and Professional Development

The project will provide protected time and project-supported opportunities for the following activities:

- **Research-software leadership:** ownership of architectural decision records, release planning, code review, incident retrospectives, maintenance estimates, and contributor onboarding, with quarterly feedback from the PI/Co-PI or independent mentor as appropriate.
- **Secure and reliable CI:** training in threat modeling, least-privilege remote execution, dependency/SBOM practices, vulnerability response, backups, and incident response. The CI Professional will participate in an external security or science-gateway review and lead the closure of at least one resulting recommendation.
- **Scientific collaboration:** monthly sessions with the science lead on model assumptions, numerical validation, uncertainty, and reproducibility; pairing with a PISM user/developer on adapter acceptance; and practice communicating engineering limitations to domain scientists.
- **Scholarship and visibility:** co-authorship or credited contributions on software/data papers and release records; at least one technical presentation, poster, tutorial, or workshop leadership role per year; and support to prepare abstracts, manuscripts, and future proposals.
- **Career development:** annual review of career pathways, promotion criteria, portfolio/CV, leadership goals, professional networks, and relevant training. Up to one focused course or conference per year may be selected within available project/institutional resources.
- **Responsible professional practice:** training in research integrity, accessibility, inclusive collaboration, software/data licensing, privacy-preserving metrics, credit and authorship, and responsible disclosure of security vulnerabilities.

Evidence and Accountability

Semiannual reviews will record completed training, new responsibilities, products, presentations, mentoring received/provided, and next objectives. The project will assess whether the individual gains independent ownership of a production component, demonstrates cross-disciplinary leadership, and has a documented advancement path. Activities and outcomes will be summarized in annual reports without disclosing private career discussions. Mentoring is a project responsibility and will not depend solely on informal availability.